Garegin Harutyunyan

C++/Qt Engineer · Real-time, cross-platform applications

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Summary

C++/Qt engineer with 4+ years building real-time, cross-platform applications. Shipped two production Qt apps with gRPC networking and encrypted streaming, led a three-engineer team, and contributed to the open-source Amnezia VPN client.

Experience

Bootech Yerevan, Armenia

C++/Qt Software Engineer May 2022 – Present

- Shipped 2 production cross-platform Qt applications delivering real-time situational data to registered users over gRPC, with authentication and encrypted streaming.
- Led a 3-engineer team through a 6-month cross-platform Qt/QML proof of concept; defined module boundaries, a shared QML component library and C++-to-QML data-flow patterns that adjacent teams reused.
- Built the company's first Qt application, a real-time information-exchange client driven by WebSocket updates; ran networking and processing on worker threads (QThread, std::thread) so the UI stayed responsive. Delivered in 6 months.

C++ Software Engineering Intern Mar 2022 – May 2022

- Designed SQLite schemas and built a C++ library exposed to Android through JNI bindings for Java/Kotlin, working in Scrum.

Freelance C++/Qt Engineer (Upwork) Feb 2024 – Jul 2025

- Contributed desktop UI features to Amnezia VPN, an open-source cross-platform VPN client: multi-panel navigation and user flows built within cross-platform constraints.
- Built a Qt Widgets image-annotation tool (QGraphicsView polygon and rectangle labeling) with OpenCV-based auto-labeling and autosave/session restore.

Armath Engineering Laboratories, Group Leader Sep 2021 – Mar 2023

- Taught programming and robotics (Python, Arduino, Raspberry Pi, intro OpenCV) to teenage students.

Skills

Languages: C++17/20, Python, SQL
Qt and libraries: Qt 5/6 (Qt Quick/QML, Widgets), gRPC, WebSockets, OpenCV, JNI
Systems: Linux, multithreading (std::thread, QThread), atomics, mutexes
Tooling: CMake, Conan, GCC/Clang, GDB, perf, Valgrind, sanitizers, Qt Quick Profiler, Git
Testing and delivery: GoogleTest/CTest, GitHub Actions, GitLab CI, Scrum
Data: SQLite, Microsoft SQL Server

Education

B.Sc. Computer Science, Yerevan State University Sep 2018 – Jun 2022

Data Structures and Algorithms, Graph Theory, Computer Architecture and Assembly, Databases, Discrete Mathematics, Linear Algebra

C++ Developer Program, Yandex Practicum Nov 2024 – Present

Advanced modern C++: STL internals, custom containers, concurrency, performance profiling

Projects

Transport Catalogue and Router (Yandex Practicum project) GitHub

- Route-planning library and CLI in C++17: builds a weighted transit graph, answers shortest-route queries (Dijkstra) over a JSON API and renders maps to SVG.
- string_view-keyed storage and preallocated containers to avoid copies; sanitizer-clean and unit-tested with GoogleTest.